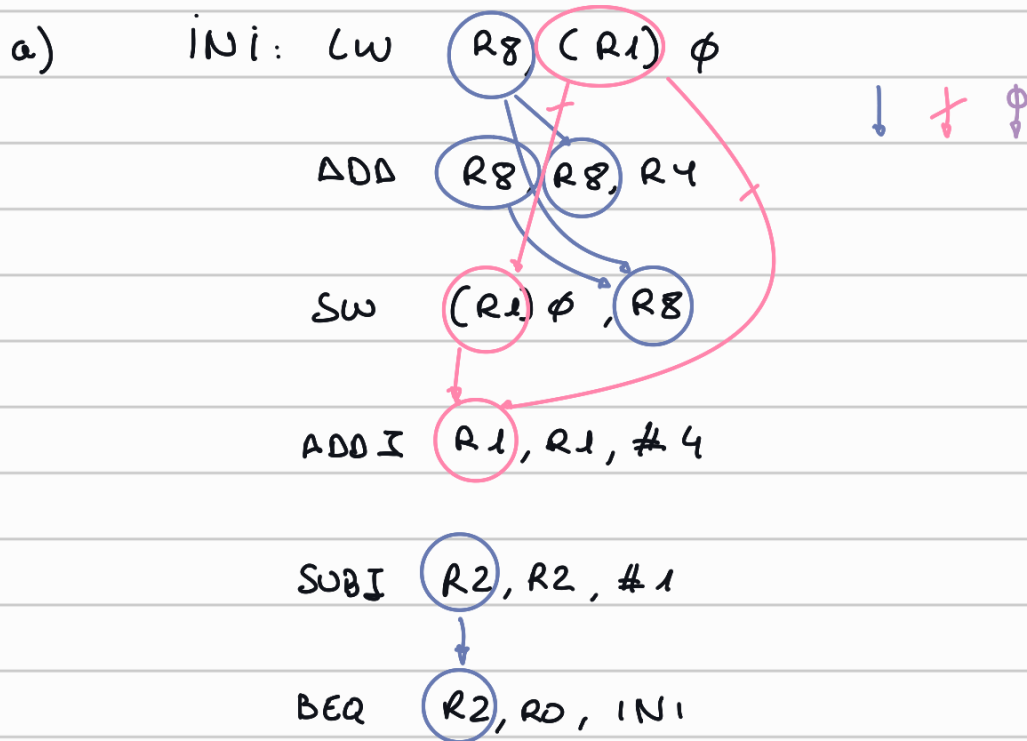


```

ini: LW   R8, (R1) 0
      ADD  R8, R8, R4
      SW   (R1) 0, R8
      ADDI R1, R1, #4 ; R1 ← R1 + 4
      SUBI R2, R2, #1  ; R2 ← R2 - 1
      BEQ  R2, R0, ini
  
```



b)

LW	R8, (R1) 0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
NOP		F	0	F	F	W																		
NOP			F	D	F	M	W																	
NOP				F	D	F	M	W																
ADD	R8, R8, R4				F	D	F	M	W															
NOP						F	D	F	M	W														
NOP							F	D	F	M	W													
NOP								F	D	F	M	W												
SW	(R1) 0, R8								F	D	F	M	W											
ADDI	R1, R1, #4									F	D	F	M	W										
SUBI	R2, R2, #1										F	D	F	M	W									
NOP												F	D	F	M	W								
NOP													F	D	F	M	W							
NOP														F	D	F	M	W						
BEQ	R2, R0, ini														F	D	F	M	W					
NOP																F	D	F	M	W				
NOP																	F	D	F	M	W			
NOP																		F	D	F	M	W		
LW	R8, (R1) 0																		F	D	F	M	W	

- 1 iteration → 19 ciclos

- 100 iterations → $22 \cdot 100 = 2200 - 3 = 2197$ ciclos

```

c)  lw 2: lw      R8, (R1) 0
    add  R8, R8, R4
    sw   (R1) 0, R8
    lw   R9, (R1) 4  Al 0ti
    add  R9, R9, R4
    sw   (R1) 4, R9
    addi R1, R1, #8  Para coger los siguientes 8
    subi R2, R2, #1  resta 1 hasta llegar a 0
    beq  R2, R0, INI-2

```

Handwritten assembly code with annotations showing register usage and value propagation:

- `Lwi2 : LW` (Registers `R8` and `(R1)` are circled)
- `ADD` (Registers `R8` and `R8` are circled; an arrow points from the `R8` in the previous instruction to the first `R8` here)
- `SW` (Registers `(R1)` and `R8` are circled; an arrow points from the `R8` in the previous instruction to the `R8` here)
- `LW` (Registers `R9` and `(R1)` are circled)
- `DDD` (Registers `R9` and `R9` are circled; an arrow points from the `R9` in the previous instruction to the first `R9` here)
- `SW` (Registers `(R1)` and `R9` are circled; an arrow points from the `R9` in the previous instruction to the `R9` here)
- `ADDI` (Registers `R1` and `R1` are circled; a pink arrow points from the `(R1)` in the previous instruction to the first `R1` here)
- `SUBI` (Registers `R2` and `R2` are circled; an arrow points from the `R2` in the previous instruction to the first `R2` here)
- `BEQ` (Registers `R2` and `R0` are circled; an arrow points from the `R2` in the previous instruction to the first `R2` here)

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
d) LW R8, (R1), 0	F	D	F	R	(W)	E																		
NOP		F			R	E																		
NOP			F		F	E																		
NOP				F	F	E																		
ADD R8, R8, R4					(D)	E		(W)																
NOP					F	E		R																
NOP						F		F																
NOP							F	D		(W)														
SW (R1), 0, R8									F	C	R	E												
LW R9, (R1), 4										(D)														
NOP										F	C	R	E											
NOP											F	C	R	E										
NOP												F	C	R	E									
DOO R9, R9, R4													(W)											
NOP														F										
NOP														F										
NOP														F										
LW (R1), 4, R9															F									
ADDI R1, R1, #8																F		(W)						
SUBI R2, R2, #1																	F	E						
NOP																		F	E					
NOP																			F	E				
NOP																				F	E			
BEQ R2, R0, IM_2																					F	E		(W)

	75	70	22	26	29	30	31	32
NOP	W							
NOP	W							
NOP	E	W						
BEQ R2, R0, IN2	D	E	W					
NOP	F	D	W					
NOP		D	E	W				
NOP			D	E	W			
NOP				D	E	W		
LW R8, (R1)0				D	E	W		

- Una iteración $\rightarrow$ 26 cicls

- 100 iteraciones $\rightarrow 26 \cdot 100 = 2600 - 3 = 2597$ cicls

e)

```

LW R8, (R1)0
LW R9, (R1)4
ADD R8, R8, R4
DOO R9, R9, R4
SUBI R2, R2, #1
SW (R1)0, R8
SW (R1)4, R9
ADDI R1, R1, #8
BEQ R2, R0, IN2
  
```

	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21
LW R8, (R1)0	F	D	F	E	W																
NOP		F	D	F	E	W															
LW R9, (R1)4			F	D	F	E	W														
NOP				F	D	F	E	W													
NOP					F	D	F	E	W												
ADD R8, R8, R4						F	D	F	E	W											
DOO R9, R9, R4							F	D	F	E	W										
NOP								F	D	F	E	W									
SUBI R2, R2, #1									F	D	F	E	W								
SW (R1)0, R8										F	D	F	E	W							
SW (R1)4, R9											F	D	F	E	W						
ADDI R1, R1, #8												F	D	F	E	W					
BEQ R2, R0, IN2													F	D	F	E	W				
NOP														F	D	F	E	W			
NOP															F	D	F	E	W		
NOP																F	D	F	E	W	
LW R8, (R1)0																	F	D	F	E	W

- Una iteración $\rightarrow$ 17 cicls

- 100 iteraciones $\rightarrow 17 \cdot 100 = 1700 - 3 = 1697$ cicls

f) Kill en beq

Anticipación $M \rightarrow Ex$ $W \rightarrow Ex$

Parados en lw - uso

BEQs sin anticipación de operandos.

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
LW $R8, (R1) \phi$	F	D	E	M	W									
ADD $R8, R8, R4$		F	D	D	E	M	W							
SW $(R1) \phi, R8$			F	D	D	E	U	W						
ADDI $R1, R1, \#4$				F	D	E	M	W						
SUBI $R2, R2, \#1$					F	D	E	M	W					
BEQ $R2, R0, IN1$						F	D	E	M	W				
...							F	-						
LW $R8, (R1) \phi$								F	D	E	M	W		

- Una iteración $\rightarrow 8$ ciclos

- 100 iteraciones $\rightarrow 11 \cdot 100 = 1100 - 3 = 1097$ ciclos

g) $Sup\ v_2 = \frac{26}{19} = 1'368$ v_2 36% más lenta

$Sup\ v_3 = \frac{17}{19} = 0'89$ v_3 11% más rápida

$Sup\ v_4 = \frac{8}{19} = 0'41$ v_4 59% más rápida